

Fiscal Policy

ECON 101 – Macroeconomics

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Chapter Outline

- 12.1 What Is Fiscal Policy?
- 12.2 The Effects of Fiscal Policy on Real GDP and the Price Level
- 12.4 Government Purchases and Tax Multipliers
- 12.5 The Limits of Fiscal Policy as a Stimulus
- 12.6 Deficits, Surpluses, and Federal Government Debt
- 12.7 The Effects of Fiscal Policy in the Long Run
- **Appendix** A Closer Look at the Multiplier

12.1 Learning Objective

Goal

Define fiscal policy and distinguish between automatic stabilizers and discretionary fiscal policy.

- **Fiscal policy:** changes in **federal government taxes and spending** intended to achieve macroeconomic objectives.
 - Provincial spending is important but primarily aimed at service delivery, not national macrostabilization.
- The federal government can **shift aggregate demand** by changing spending and tax decisions.
- **Canada 2020 example:** the federal government launched \$314 billion in emergency spending (CERB, CEWS, CERS) — the largest fiscal intervention in Canadian history.

Automatic Stabilizers vs. Discretionary Fiscal Policy

Automatic Stabilizers

Spending and taxes that automatically change with the business cycle **without new legislation**.

- Employment Insurance (EI): rises in recessions as more workers qualify.
- Income tax: falls in recessions (lower incomes $\Rightarrow$ lower tax revenue).
- Social assistance: rises when unemployment increases.

Act fast, no legislative delay.

Discretionary Fiscal Policy

Intentional government actions requiring new legislation.

- 2009 Economic Action Plan: \$40B infrastructure stimulus.
- 2020 CERB: \$2,000/month to unemployed Canadians.
- 2024 GST holiday: temporary sales tax cut.
- 2025 income tax cut proposal.

Takes time; subject to political process.

12.2 Learning Objective

Goal

Explain how fiscal policy affects aggregate demand and how it can stabilize the economy.

- **Two channels:**
 - Changes in government purchases (G): affects AD **directly**.
 - Changes in taxes (T): affects disposable income $\Rightarrow$ consumption $\Rightarrow$ AD **indirectly**.
- **Expansionary fiscal policy:** $\uparrow G$ or $\downarrow T \Rightarrow$ AD shifts right.
- **Contractionary fiscal policy:** $\downarrow G$ or $\uparrow T \Rightarrow$ AD shifts left.

Figure 12.1 – Expansionary Fiscal Policy (Closing a Recessionary Gap)

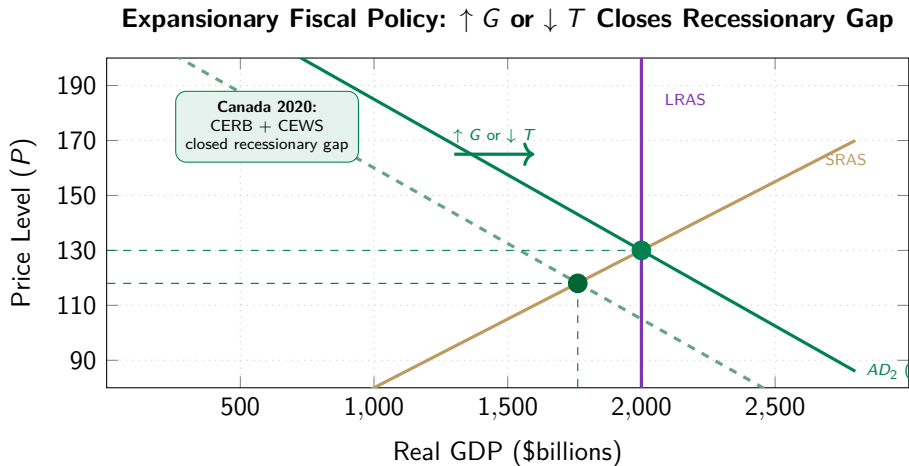


Figure 12.2 – Contractionary Fiscal Policy (Fighting Inflation)

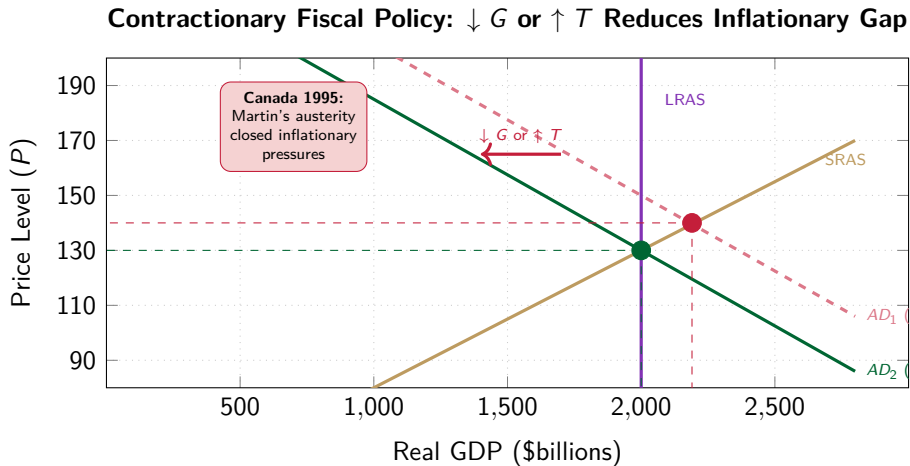


Table 12.1 – Countercyclical Fiscal Policy in Canada

Problem	Policy Type	Government Action	Canadian Example
Recession	Expansionary	$\uparrow G$ or $\downarrow T \Rightarrow$ AD shifts right	CERB/CEWS (\$314B) 2020
Rising inflation	Contractionary	$\downarrow G$ or $\uparrow T \Rightarrow$ AD shifts left	1995 Martin budget cuts

- **Note:** in Table 12.1 the original text references the U.S. Congress. The Canadian equivalent is Parliament — specifically the House of Commons, which must pass all spending bills and tax changes.
- Countercyclical policy aims to **dampen** the business cycle, not eliminate it.

12.4 Learning Objective

Goal

Explain how the government purchases and tax multipliers work.

- A \$1 increase in government spending raises equilibrium GDP by **more than \$1**.
- Why?
 - Initial spending $\Rightarrow$ incomes rise $\Rightarrow$ consumption rises $\Rightarrow$ incomes rise further $\Rightarrow$...
 - This chain is the **multiplier effect**.
- **Government purchases multiplier:**

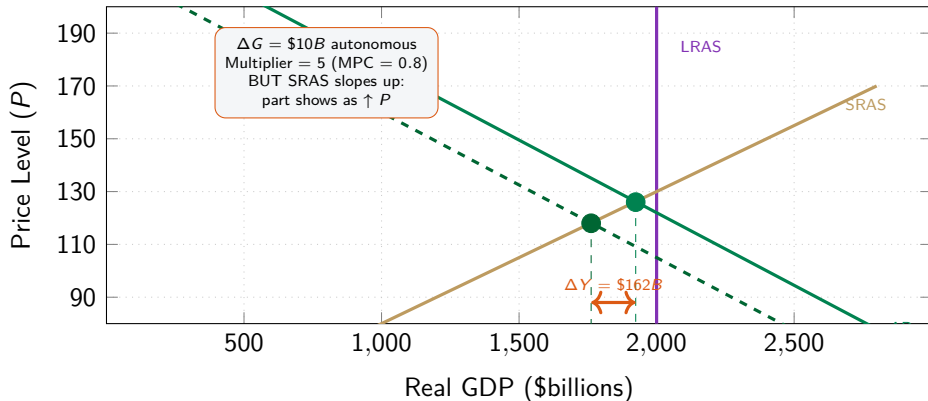
$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - \text{MPC}}.$$

- **Tax multiplier:**

$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1 - \text{MPC}}.$$

Figure 12.3 – The Multiplier Effect in AD-AS

Multiplier: \$10B Increase in G Raises Equilibrium GDP by \$50B ($MPC = 0.8$)



Government Purchases Multiplier vs. Tax Multiplier

Government Purchases Multiplier

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - \text{MPC}}$$

- \$1 of G **directly** adds \$1 to AD.
- Then the induced consumption chain kicks in.
- With $\text{MPC} = 0.8$: multiplier = 5.
- $\Delta G = \$10B \Rightarrow \Delta Y = \$50B$.

Tax Multiplier

$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1 - \text{MPC}}$$

- Tax cut raises disposable income.
 - Households **save MPS** portion, spend MPC portion.
 - First-round effect: $\$ \text{MPC}$ (not \$1).
 - With $\text{MPC} = 0.8$: tax multiplier = -4 .
 - $\Delta T = -\$10B \Rightarrow \Delta Y = +\$40B$.
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- The **government purchases multiplier is larger** (absolute value) than the tax multiplier: spending injection is fully spent, while a tax cut is partly saved.

12.5 Learning Objective

Goal

Discuss the difficulties that can arise in implementing fiscal policy.

- Fiscal policy is powerful in theory but faces practical limitations:
 1. **Timing problems** (legislative and implementation lags).
 2. **Crowding out** (higher government borrowing raises interest rates, reducing private spending).
 3. **Political constraints** (fiscal policy is a political process, not just an economic one).

Timing Problems

1. Recognition lag:

- GDP data arrives 2–3 months after the quarter ends. By the time a recession is recognized, it may already be deep.

2. Legislative delay:

- Parliament must debate and approve spending or tax changes.
- **Canadian example (2009):** Conservative minority government's Economic Action Plan took months to implement after the recession had begun.

3. Implementation delay:

- Infrastructure projects take months to years to actually begin (design, tender, construction).
- **“Shovel-ready” projects:** a political priority in 2009 precisely because of this lag.

4. Because of these delays, fiscal stimulus may arrive **after the recession has ended** — adding fuel to the next boom.

Crowding Out

Crowding Out

A decline in private expenditures (C, I, NX) that results from an increase in government purchases.

- **Mechanism:**

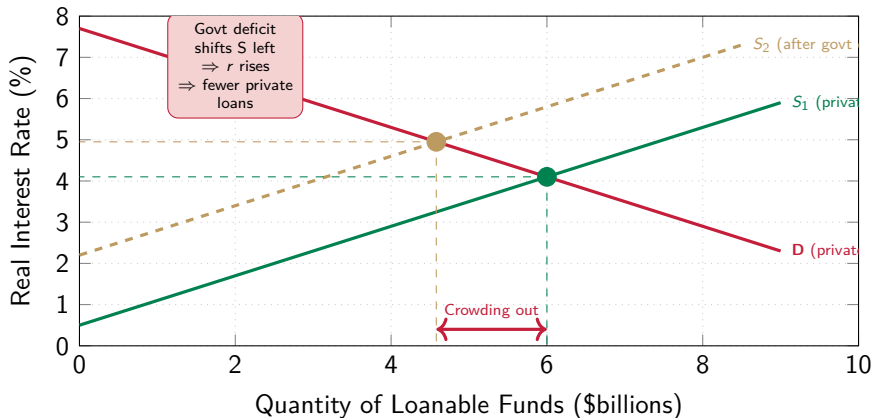
- Government increases spending $\Rightarrow$ borrows more (issues more bonds).
- Higher bond supply $\Rightarrow$ bond prices fall $\Rightarrow$ interest rates rise.
- Higher interest rates $\Rightarrow$ lower private investment, lower consumer credit, stronger CAD (lower NX).
- Private spending falls — partially offsetting the stimulus.

- **Canadian context:**

- Canada's federal debt is $\approx$ \$1.3 trillion (2024).
- Higher debt-to-GDP ratios may gradually raise borrowing costs.
- But with BoC independence, crowding out is partially offset by monetary policy.

Figure 12.4 – Crowding Out in the Loanable Funds Market

Government Borrowing Raises Interest Rates, Crowding Out Private Investment



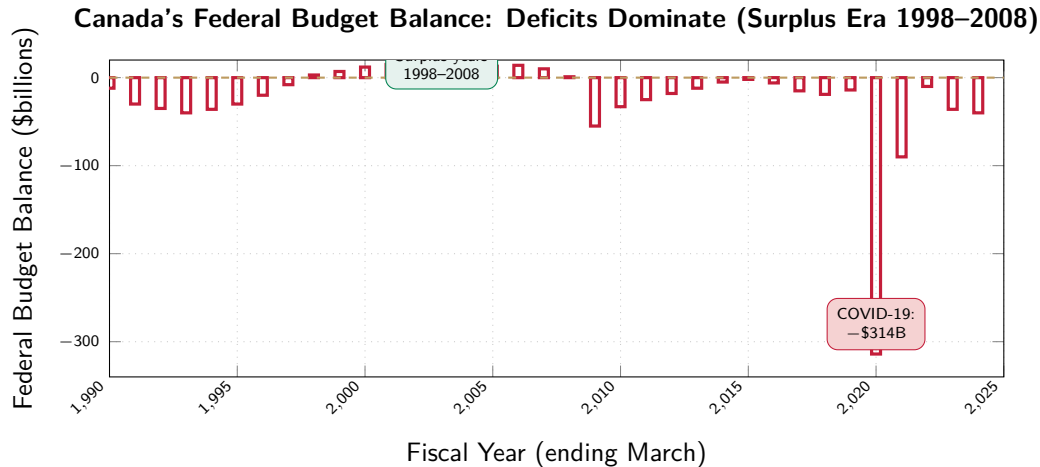
12.6 Learning Objective

Goal

Define budget deficit and federal government debt, and explain how the budget serves as an automatic stabilizer.

- **Budget deficit:** expenditures $>$ tax revenues.
- **Budget surplus:** expenditures $<$ tax revenues.
- **Federal government debt (national debt):** the total stock of outstanding Government of Canada bonds.
- **Canadian fiscal history:**
 - Deficits almost every year since 1970.
 - One stretch of surpluses: 1998–2008 (Chrétien/Martin era).
 - **2020–21:** deficit of \$314 billion — by far the largest ever.
 - **2024–25:** deficit $\approx$ \$40 billion.

Figure 12.5 – Canada's Federal Budget Balance (1990–2024)



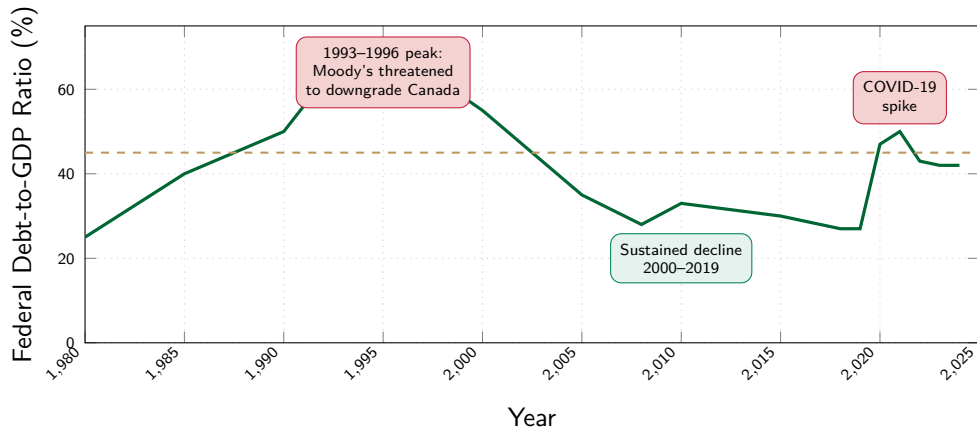
Source: Department of Finance Canada, Federal Budget. Positive = surplus; negative = deficit.

Automatic Stabilizers and the Business Cycle

- **In recessions** (automatic stabilizers kick in):
 - Tax revenues **fall** (lower incomes, profits, consumption).
 - EI payments, social assistance **rise**.
 - Result: budget deficit grows automatically, helping support AD.
- **In expansions:**
 - Tax revenues **rise**.
 - EI, welfare payments **fall**.
 - Result: budget moves toward surplus automatically, dampening inflation.
- **Canada's EI example (2020):**
 - EI and emergency benefits paid out $\approx$ \$100 billion in 2020 alone.
 - This replaced lost income for millions of Canadians, supporting consumption.
 - Without this, the recession would have been **much deeper**.

Figure 12.6 – Federal Government Debt to GDP Ratio (Canada)

Federal Debt-to-GDP Rose Sharply in 1990s, Then Fell, Then Rose Again (2020)



Source: Department of Finance Canada; IMF Fiscal Monitor. G7 average is approximate.

Is Government Debt a Problem?

- **Short-term view:** Canada's federal debt-to-GDP ratio ($\approx 42\%$) is lower than most G7 countries. Not an immediate crisis.
- **Long-term concerns:**
 - If debt grows faster than GDP, interest payments consume a larger share of the budget.
 - **Crowding out:** rising debt may gradually push up real interest rates, reducing private investment.
 - **Generational equity:** future taxpayers (today's children) inherit the debt.
- **When is debt less problematic?**
 - When it finances **productive investments**: infrastructure, education, R&D.
 - These raise future potential GDP, making the debt easier to service.
 - **Canada 2020:** emergency spending was necessary to prevent economic collapse, even though it was largely consumption-focused.

12.7 Learning Objective

Goal

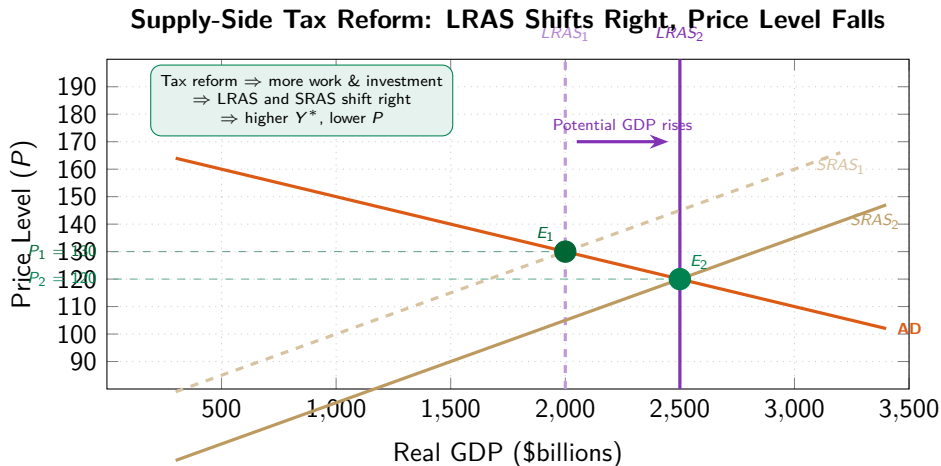
Discuss the effects of fiscal policy in the long run (supply-side effects).

- So far: fiscal policy as a short-run stabilization tool (shifts AD).
- Now: how fiscal policy affects **potential GDP** (shifts LRAS).
- **Supply-side economics:**
 - Fiscal policy actions aimed at increasing long-run aggregate supply.
 - Focus: taxes and regulations that affect incentives to work, save, invest, and innovate.

The Tax Wedge and Incentives

- Most taxes create a **wedge** between the pre-tax and post-tax return to an activity.
- **Examples:**
 - **Personal income tax:** if your marginal rate is 53% (federal + BC combined), each extra dollar earned yields only 47 cents after tax.
 - May reduce hours worked, entrepreneurship, or human capital investment.
 - **Corporate income tax:** Canada's combined rate $\approx 26\%$ in 2024.
 - High rates may reduce investment in new machinery, R&D, and job creation.
 - **Capital gains tax:** affects saving and investment decisions.
 - **2024 Canada:** Trudeau government raised capital gains inclusion rate from 50% to 67% — controversial impact on investment incentives.

Figure 12.7 – Supply-Side Effects of a Tax Cut



Supply-side tax reform raises potential GDP from \$2,000B to \$2,500B while lowering the long-run price level.

Long-Run Effects of Tax Reform: What We Know

- Tax reform can increase potential GDP in the long run by:
 - Raising labour supply (more hours worked, more participation).
 - Increasing saving and investment (more capital formation).
 - Encouraging entrepreneurship and innovation.
- **Uncertainty:**
 - Magnitude of supply-side effects is **empirically contested**.
 - People may want to work more at lower marginal rates, but may face *constraints* (fixed schedules, childcare).
 - Tax cuts also create short-run demand effects (shifting AD right), which may dominate.
- **Canadian policy debate:**
 - 2024 capital gains inclusion rate increase: critics argued it would deter investment; supporters argued revenue would fund social priorities.
 - Active empirical research ongoing.

Appendix – The Government Purchases and Tax Multipliers

- Simple Canadian macroeconomic model (closed economy, fixed price level):

$$C = 1,000 + 0.8(Y - T), \quad T = 250, \\ I = 300, \quad G = 500.$$

- Equilibrium:** $Y = C + I + G$:

$$Y = 1,000 + 0.8(Y - 250) + 300 + 500.$$

$$Y = 1,600 + 0.8Y - 200 = 1,400 + 0.8Y.$$

$$0.2Y = 1,400 \Rightarrow Y = \$7,000 \text{ billion.}$$

- Government purchases multiplier:** $\frac{\Delta Y}{\Delta G} = \frac{1}{1-0.8} = 5.$
- $\Delta G = +\$10B \Rightarrow \Delta Y = +\$50B.$

The Tax Multiplier and the Balanced Budget Multiplier

- **Tax multiplier:**

$$\frac{\Delta Y}{\Delta T} = -\frac{\text{MPC}}{1 - \text{MPC}} = -\frac{0.8}{0.2} = -4.$$

- $\Delta T = -\$10B$ (tax cut) $\Rightarrow \Delta Y = +\$40B$.
- **Why smaller than G multiplier?** A \$10B tax cut gives \$10B to households, but they **save** $0.2 \times 10 = \$2B$, so only \$8B is initially spent.
- **Balanced budget multiplier:**
 - Increase G and T by the same amount (\$10B).
 - $\Delta Y = (+5) \times 10B + (-4) \times 10B = \$10B$.
 - Balanced budget multiplier = $\frac{\Delta Y}{\Delta G} = 1$.
 - Equal dollar increases in spending and taxes raise GDP by the same dollar.

Summary

- **Fiscal policy:** changes in federal spending (G) and taxes (T) to achieve macroeconomic goals.
- **Automatic stabilizers** (EI, progressive income tax) cushion downturns without legislation.
- **Expansionary** ($\uparrow G$ or $\downarrow T$) closes recessionary gaps; **contractionary** ($\downarrow G$ or $\uparrow T$) reduces inflationary pressure.
- Government purchases multiplier = $\frac{1}{1-MPC}$; tax multiplier = $-\frac{MPC}{1-MPC}$.
- Practical limits: **timing lags** (recognition, legislative, implementation) and **crowding out**.
- Canada's federal debt-to-GDP $\approx 42\%$ (2024) — manageable but trending higher.
- **Supply-side** fiscal policy (tax reform) can shift LRAS right and raise potential GDP, but magnitudes are uncertain.
- **Balanced budget multiplier** = 1: equal increases in G and T raise GDP by the same dollar.
- **Canada 2020 case study:** \$314B in emergency spending was the largest ever fiscal expansion — essential to prevent depression-level output losses, at the cost of a large deficit and higher debt.

Thank you!

Questions?